

Exponential vs. Linear Growth — Solutions

Full Step-by-Step | Trap Alerts | Key Insights | Fast Methods | Desmos Verdicts

Q1

Easy

Answer: B) $A(t) = 500 + 40t$

Step-by-Step Solution:

The account earns a fixed \$40 per year (constant rate of change).

This is linear growth.

Linear model: $A(t) = \text{initial} + \text{rate} \times t = 500 + 40t$.

Why Other Options Are Wrong:

A) This would be 8% compounding — exponential, not linear.

C) and D) These have nonsensical exponential structures.

TRAP ALERT

\$40 per year is a constant amount, which means linear. If it were 8% per year (a percentage), it would be exponential.

KEY INSIGHT

Constant AMOUNT added = linear. Constant PERCENTAGE change = exponential. This is the single most important distinction.

FAST METHOD

Fixed dollar amount = linear = $mx + b$ form. $A(t) = 500 + 40t$.

VERDICT

Hand-solve. Identify the model type — 5 seconds.

Q2

Easy

Answer: B) $P(t) = 200(2)^{t/3}$

Step-by-Step Solution:

Doubles every 3 hours = exponential growth with base 2 and period 3.

After t hours, the number of 3-hour periods is $t/3$.

$$P(t) = 200(2)^{t/3}.$$

Check: at $t=3$, $P = 200(2)^1 = 400$ (doubled). Correct!

Why Other Options Are Wrong:

A) Linear model — bacteria do not grow by a fixed amount.

C) Uses base 3 and divides exponent by 2 — wrong base and wrong period.

D) Uses $3t$ in the exponent — would triple every hour, and the exponent is wrong.

TRAP ALERT

The exponent must be $t/3$ (dividing time by the doubling period), not $3t$. Students often put the period in the wrong place.

KEY INSIGHT

If something multiplies by factor f every p time units, the function is: $\text{initial} \times f^{(t/p)}$.

FAST METHOD

Doubles ($f=2$) every 3 hours ($p=3$). Exponent = $t/3$. $P = 200(2)^{t/3}$.

VERDICT

Hand-solve. Formula construction — 10 seconds.

Q3

Easy

Answer: D) Exponential growth, constant ratios

Step-by-Step Solution:

Check differences: 60, 63.60, 67.42 — NOT constant (not linear).

Check ratios: $1060/1000 = 1.06$, $1123.60/1060 = 1.06$, $1191.02/1123.60 = 1.06$.

Ratios are constant (1.06), confirming exponential growth at 6% per year.

Why Other Options Are Wrong:

- A) Differences are NOT constant (60, 63.60, 67.42).
- B) Linear growth has constant differences, not constant ratios.
- C) Exponential has constant ratios, not constant differences.

TRAP ALERT

Students check differences and see numbers close to 60, thinking they are "close enough" to be constant. Always check ratios too — if ratios are exactly constant, it is exponential.

KEY INSIGHT

Linear: constant first differences. Exponential: constant ratios between consecutive terms. This is the fundamental test.

FAST METHOD

Divide consecutive terms: all give 1.06. Constant ratio = exponential.

VERDICT

Hand-solve. Two quick divisions — 15 seconds.

Q4

Easy

Answer: B) $f(x) = 7(3)^x$

Step-by-Step Solution:

An exponential function has the form $f(x) = a(b)^x$, where the variable x is in the exponent.

$f(x) = 7(3)^x$ has x in the exponent. This is exponential.

Why Other Options Are Wrong:

- A) $f(x) = 3x + 7$ is linear.
- C) $f(x) = 3x^2 + 7$ is quadratic (power function).
- D) $f(x) = 7x^3$ is cubic (power function).

TRAP ALERT

Students confuse exponential (variable in the exponent) with power functions (variable in the base). x^2 is a power function; 2^x is exponential.

KEY INSIGHT

Exponential: $f(x) = a(b)^x$ (variable is the exponent). Power: $f(x) = ax^n$ (variable is the base).

FAST METHOD

Look for the variable in the exponent position. Only option B has this.

VERDICT

Hand-solve. Pattern recognition — 5 seconds.

Q5

Easy

Answer: B) $24000(0.85)^5$

Step-by-Step Solution:

Depreciates by 15% per year means it retains $85\% = 0.85$ each year.

After 5 years: $V = 24000(0.85)^5$.

Why Other Options Are Wrong:

A) 0.15 is the decay rate, not the retention rate. This would shrink the value to nearly zero.

C) This is linear depreciation, not percentage-based.

D) 1.15 means 15% growth, not depreciation.

TRAP ALERT

Students use 0.15 (the percentage lost) instead of 0.85 (the percentage retained). The base must be $(1 - \text{rate})$ for decay.

KEY INSIGHT

For percent decrease: multiply by $(1 - \text{rate})$. For percent increase: multiply by $(1 + \text{rate})$.

FAST METHOD

15% loss $\rightarrow$ retain 85% $\rightarrow$ base = 0.85. $V = 24000(0.85)^5$.

VERDICT

Hand-solve. Identify the base — 10 seconds. Use Desmos only if you need the numerical value.

Q6

Easy

Answer: 36

Step-by-Step Solution:

Linear growth: height = initial + rate $\times$ time.

Height = $12 + 3(8) = 12 + 24 = 36$ cm.

TRAP ALERT

Grid-in: straightforward, but students might multiply $12 \times 3 \times 8 = 288$ or use an exponential model. 3 cm per week is constant = linear.

KEY INSIGHT

Constant rate of change = linear. height = $12 + 3t$.

FAST METHOD

$12 + 24 = 36$.

VERDICT

Hand-solve. Arithmetic — 5 seconds.

Q7

Medium

Answer: B) Loses 8% per year

Step-by-Step Solution:

The base is $0.92 = 1 - 0.08$.

This means the equipment retains 92% of its value each year.

It loses 8% per year.

Why Other Options Are Wrong:

A) It retains 92%, not loses 92%.

C) $0.92 < 1$ means decay (loss), not growth.

D) The loss is a percentage (8%), not a fixed dollar amount.

TRAP ALERT

Students confuse the base ($0.92 =$ retention rate) with the rate of loss ($0.08 = 8\%$). The DSAT loves testing whether students can interpret the base of an exponential function.

KEY INSIGHT

If base $b < 1$: decay rate $= 1 - b$. If base $b > 1$: growth rate $= b - 1$.

FAST METHOD

$1 - 0.92 = 0.08 = 8\%$ loss.

VERDICT

Hand-solve. Interpretation — 5 seconds.

Q8

Medium

Answer: B) 2022

Step-by-Step Solution:

Town (linear): $P_T = 12000 + 800t$, where $t =$ years after 2015.

City (exponential): $P_C = 8000(1.12)^t$.

Test values:

$t=5$ (2020): Town = 16,000. City $= 8000(1.12)^5 = 8000(1.7623) = 14,098$. Town leads.

$t=6$ (2021): Town = 16,800. City $= 8000(1.12)^6 = 8000(1.9738) = 15,790$. Town still leads.

$t=7$ (2022): Town = 17,600. City $= 8000(1.12)^7 = 8000(2.2107) = 17,686$. City exceeds town!

The city first exceeds the town in 2022 (at $t = 7$).

Why Other Options Are Wrong:

A) 2020: City is still below ($14,098 < 16,000$).

C) 2024: City already exceeded the town two years earlier.

D) 2026: Much too late.

TRAP ALERT

The exponential growth catches up sooner than expected. Students who guess without testing values might overshoot to 2024 or 2026.

KEY INSIGHT

Exponential growth always eventually surpasses linear growth, but you need to test specific values to find exactly when.

FAST METHOD

Use Desmos: plot $y = 12000 + 800x$ and $y = 8000(1.12)^x$. Find the intersection at approximately $x = 7$ (year 2022).

VERDICT

Desmos is faster. Plotting both functions and finding the intersection visually takes 20 seconds vs. manual trial-and-error.

Q9

Medium

Answer: B) Number of subscribers in January

Step-by-Step Solution:

In $S(t) = 250(1.06)^t$, when $t = 0$: $S(0) = 250(1.06)^0 = 250(1) = 250$.

The 250 is the initial value — the number of subscribers at $t = 0$ (January).

Why Other Options Are Wrong:

A) The growth rate is 6% (from the 1.06), not 250.

C) $S(6) = 250(1.06)^6 = \text{about } 355$, not 250.

D) The monthly gain is not constant (it is 6% of the current count, which changes).

TRAP ALERT

The initial value (coefficient) and the growth rate (base) are frequently confused. The DSAT tests this interpretation repeatedly.

KEY INSIGHT

In $f(t) = a(b)^t$: a is the initial value (when $t = 0$), b is the growth/decay factor, and $b - 1$ is the rate.

FAST METHOD

Set $t = 0$: $250 \times 1 = 250 = \text{initial subscribers}$.

VERDICT

Hand-solve. Interpretation — 5 seconds.

Q10

Medium

Answer: A) 8

Step-by-Step Solution:

Culture 1 (exponential): $C1(t) = 500(3)^{t/4}$.

Culture 2 (linear): $C2(t) = 2000 + 200t$.

$t = 4$: $C1 = 500(3)^1 = 1,500$. $C2 = 2000 + 800 = 2,800$. $C2$ leads.

$t = 6$: $C1 = 500(3)^{1.5} = 500 \times 5.196 = 2,598$. $C2 = 2000 + 1200 = 3,200$. $C2$ still leads.

$t = 7$: $C1 = 500(3)^{1.75} = 500 \times 6.84 = 3,420$. $C2 = 2000 + 1400 = 3,400$. $C1$ barely exceeds $C2$!

$t = 8$: $C1 = 500(3)^2 = 4,500$. $C2 = 2000 + 1600 = 3,600$. $C1$ clearly exceeds.

The crossover happens between $t = 6$ and $t = 7$. The first listed option where $C1$ exceeds $C2$ is approximately 8 hours.

Why Other Options Are Wrong:

B) 10 — $C1$ already exceeded $C2$ before hour 8.

C) 12 — much too late; $C1$ is far ahead by then.

D) 16 — far too late.

TRAP ALERT

The exponential culture starts much smaller (500 vs 2,000) but catches up due to tripling. Students must test specific values rather than guessing.

KEY INSIGHT

When comparing exponential and linear growth, the exponential always wins eventually. Test values at doubling/tripling points ($t = 4, 8, 12\dots$) for clean numbers.

FAST METHOD

Desmos: plot $y = 500(3)^{(x/4)}$ and $y = 2000 + 200x$. Find intersection near $x = 7$.

VERDICT

Desmos is faster. Plotting both curves and finding the intersection takes 20 seconds.

Q11

Medium

Answer: 51.2

Step-by-Step Solution:

Decays by 20% every 6 years = retains 80% = 0.80 every 6 years.

18 years = 3 periods of 6 years.

Mass = $100(0.80)^3 = 100(0.512) = 51.2$ grams.

TRAP ALERT

Students might use 0.20 (the decay rate) instead of 0.80 (the retention rate), giving $100(0.2)^3 = 0.8$ grams — way too small.

KEY INSIGHT

Decay by $r\%$ means retention of $(100 - r)\%$. Use $(1 - r)$ as the base, not r .

FAST METHOD

3 halvings of 80%: $100 \times 0.8 \times 0.8 \times 0.8 = 51.2$.

VERDICT

Hand-solve. Three simple multiplications — 15 seconds.

Q12

Medium

Answer: B) $y = 3, 9, 27, 81$

Step-by-Step Solution:

Check ratios between consecutive y -values:

Option A: $10/5=2$, $15/10=1.5$. Not constant $\rightarrow$ not exponential.

Option B: $9/3=3$, $27/9=3$, $81/27=3$. Constant ratio of 3 $\rightarrow$ exponential!

Option C: $5/2=2.5$, $10/5=2$. Not constant.

Option D: $8/4=2$, $14/8=1.75$. Not constant.

Why Other Options Are Wrong:

A) Constant differences (linear).

C) Neither constant differences nor constant ratios.

D) Neither constant differences nor constant ratios.

TRAP ALERT

Option A has constant differences (5), making it linear, which students might confuse with exponential.

KEY INSIGHT

Exponential = constant ratios. Linear = constant differences. Always check ratios of consecutive terms.

FAST METHOD

Divide consecutive y-values. If all quotients are equal, it is exponential.

VERDICT

Hand-solve. Quick division check — 15 seconds.

Q13

Medium

Answer: C) After approximately 13 years

Step-by-Step Solution:

Card 1 (exponential): $V1 = 60(1.05)^t$.

Card 2 (linear): $V2 = 60 + 4t$.

Initially, exponential growth = 5% of \$60 = \$3/year, which is LESS than the linear \$4/year.

So the linear card starts ahead. But exponential growth accelerates over time.

Test values:

$t = 10$: $V1 = 60(1.05)^{10} = \$97.73$. $V2 = 60 + 40 = \$100$. Linear still leads.

$t = 12$: $V1 = 60(1.05)^{12} = \$107.75$. $V2 = 60 + 48 = \$108$. Nearly equal!

$t = 13$: $V1 = 60(1.05)^{13} = \$113.14$. $V2 = 60 + 52 = \$112$. Exponential just passed!

The first card first exceeds the second after approximately 13 years.

Why Other Options Are Wrong:

A) The linear card earns more per year initially (\$4 vs \$3), so the exponential is NOT always worth more.

B) Exponential always eventually outpaces linear — so the first card WILL eventually be worth more.

D) 25 years is well past the crossover point (crossover is at year 13).

TRAP ALERT

The linear card initially grows faster (\$4/year $>$ 5% of \$60 = \$3/year). Students might assume the exponential always leads, but it starts behind and must catch up.

KEY INSIGHT

When the exponential rate produces less than the linear amount initially, the crossover takes longer. But exponential always wins in the long run.

FAST METHOD

Desmos: plot $y = 60(1.05)^x$ and $y = 60 + 4x$. Find the intersection at approximately $x = 13$.

VERDICT

Desmos is much faster. Plotting reveals the crossover in seconds vs. tedious manual computation.

Q14

Medium

Answer: C) $t = 8$

Step-by-Step Solution:

$f(t) = 50(1.40)^t$ (exponential, starts at 50).

$g(t) = 200 + 60t$ (linear, starts at 200).

The exponential starts much lower but grows by 40% per month.

Test values:

$t = 4$: $f = 50(1.40)^4 = 192.1$. $g = 200 + 240 = 440$. g leads by far.

$t = 6$: $f = 50(1.40)^6 = 376.5$. $g = 200 + 360 = 560$. g still leads.

$t = 8$: $f = 50(1.40)^8 = 737.9$. $g = 200 + 480 = 680$. f exceeds g !

$t = 7$: $f = 527.1$. $g = 620$. g still leads.

$f(t)$ first exceeds $g(t)$ at $t = 8$.

Why Other Options Are Wrong:

A) $t = 4$: $f = 192$ vs $g = 440$. g leads by a wide margin.

B) $t = 6$: $f = 377$ vs $g = 560$. g still leads.

D) $t = 10$: f already exceeded g two months earlier.

TRAP ALERT

The exponential starts at only 50 while the linear starts at 200, so students may think the exponential never catches up. But 40% growth is rapid and eventually dominates.

KEY INSIGHT

Test the answer choices by substitution: compute $f(t)$ and $g(t)$ for each option and see where f first exceeds g .

FAST METHOD

Desmos: plot $y = 50(1.4)^x$ and $y = 200 + 60x$. Find the intersection near $x = 8$.

VERDICT

Desmos is ideal. Graph both functions and visually locate where the exponential curve crosses above the line.

Q15

Medium

Answer: A) 3

Step-by-Step Solution:

$f(0) = a(b)^0 = a = 5$. So $a = 5$.

$f(2) = 5(b)^2 = 45$.

$b^2 = 45/5 = 9$.

$b = 3$ (taking the positive root since $b > 0$ for exponential growth).

Why Other Options Are Wrong:

- B) 5 — confuses a and b .
- C) 9 — gives b^2 , not b .
- D) 40 — computes $45 - 5 = 40$.

TRAP ALERT

Option C (9) is the squared value trap. Students solve $b^2 = 9$ but forget to take the square root.

KEY INSIGHT

Use $(0, y)$ to find a (since $b^0 = 1$). Then use the second point to find b .

FAST METHOD

$a = 5$ (from the y -intercept). $b^2 = 9 \rightarrow b = 3$.

VERDICT

Hand-solve. Two quick steps — 15 seconds.

Q16

Hard

Answer: A) $A(t) = 600(0.5)^{t/4}$

Step-by-Step Solution:

Half-life = 4 hours means the amount is halved every 4 hours.

After t hours, the number of half-life periods is $t/4$.

$$A(t) = 600(0.5)^{t/4}.$$

Check: at $t=4$, $A = 600(0.5)^1 = 300$ (half of 600). Correct!

Why Other Options Are Wrong:

- B) Uses $4t$ — would halve every $1/4$ hour, way too fast.
- C) Linear model — does not capture exponential decay.
- D) Uses $4/t$ — the exponent should be $t/4$, not $4/t$.

TRAP ALERT

The exponent is $t/(\text{half-life})$, not $(\text{half-life}) \times t$. This mirrors the doubling-time formula in Q2.

KEY INSIGHT

Half-life formula: $A(t) = A_0(0.5)^{t/\text{half-life}}$. This is a special case of exponential decay.

FAST METHOD

Half-life = 4. Exponent = $t/4$. $A = 600(0.5)^{t/4}$.

VERDICT

Hand-solve. Formula recall — 10 seconds.

Q17

Hard

Answer: 10

Step-by-Step Solution:

$$f(t) = 2000(0.97)^t < 1500.$$

$$(0.97)^t < 1500/2000 = 0.75.$$

Take $\ln$ of both sides: $t \times \ln(0.97) < \ln(0.75)$.

$$\ln(0.97) = -0.03046, \ln(0.75) = -0.2877.$$

$$t > 0.2877/0.03046 = 9.44\dots$$

After 9.44 days, so the first complete day is day 10.

$$\text{Check: } f(9) = 2000(0.97)^9 = 2000(0.7602) = 1520.4 > 1500.$$

$$f(10) = 2000(0.97)^{10} = 2000(0.7374) = 1474.8 < 1500.$$

First drops below 1500 after 10 complete days.

TRAP ALERT

Grid-in: the question asks for complete days. $f(9)$ is still above 1500, but $f(10)$ is below. The answer is 10, not 9.

KEY INSIGHT

Solve exponential inequalities using logarithms. Remember to flip the inequality when dividing by a negative number ($\ln$ of a number less than 1 is negative).

FAST METHOD

Desmos: graph $y = 2000(0.97)^x$ and $y = 1500$. Find where they intersect. x is about 9.4, so answer is 10.

VERDICT

Desmos is much faster. Graphing and finding the intersection takes 15 seconds vs. manual logarithm work.

Q18

Hard

Answer: B) Year 8

Step-by-Step Solution:

Plan A (linear): $S_A = 50000 + 2000t$.

Plan B (exponential): $S_B = 45000(1.05)^t$.

Test values:

$$t = 6: A = 50000 + 12000 = \$62,000. B = 45000(1.05)^6 = 45000(1.3401) = \$60,305. A \text{ leads.}$$

$$t = 7: A = 50000 + 14000 = \$64,000. B = 45000(1.05)^7 = 45000(1.4071) = \$63,320. A \text{ still leads.}$$

$$t = 8: A = 50000 + 16000 = \$66,000. B = 45000(1.05)^8 = 45000(1.4775) = \$66,486. B \text{ exceeds A!}$$

Plan B first offers a higher salary in Year 8.

Why Other Options Are Wrong:

A) Year 6: Plan A still leads (\$62,000 vs \$60,305).

C) Year 10: Plan B already exceeded Plan A two years earlier.

D) Year 12: Much too late.

TRAP ALERT

The exponential plan starts \$5,000 lower, so students may think it takes a very long time to catch up. But 5% compounding accelerates quickly.

KEY INSIGHT

Exponential salary growth always eventually beats linear raises. The crossover depends on the specific starting values and rates.

FAST METHOD

Desmos: plot $y = 50000 + 2000x$ and $y = 45000(1.05)^x$. Find intersection at approximately $x = 8$.

VERDICT

Desmos is ideal. Graphing both functions reveals the crossover in seconds.

Q19

Hard

Answer: B) Horizontal asymptote at $y = -5$

Step-by-Step Solution:

$$g(x) = 3(2)^x - 5.$$

As x approaches negative infinity, 2^x approaches 0.

So $g(x)$ approaches $3(0) - 5 = -5$.

The horizontal asymptote is $y = -5$.

Why Other Options Are Wrong:

A) $y = 0$ is the asymptote for $f(x) = 3(2)^x$ without the -5 shift.

C) Exponential functions have horizontal (not vertical) asymptotes.

D) All exponential functions have exactly one horizontal asymptote.

TRAP ALERT

Students might forget the vertical shift. The -5 moves the asymptote from $y=0$ down to $y=-5$.

KEY INSIGHT

For $f(x) = a(b)^x + c$, the horizontal asymptote is $y = c$. The constant c shifts the baseline.

FAST METHOD

The $+c$ term (here, -5) is the horizontal asymptote.

VERDICT

Hand-solve. Read off the vertical shift — 5 seconds.

Q20

Hard

Answer: B) City A will eventually exceed City B

Step-by-Step Solution:

City A: exponential growth at 3%/year.

City B: linear growth of 1,200/year.

Exponential growth always eventually outpaces linear growth, regardless of starting values.

City A will eventually surpass City B.

Why Other Options Are Wrong:

A) This is only true in the short term.

C) They will cross, and then A keeps growing faster — they do not stay equal.

D) Linear growth never decreases (it adds 1,200 per year).

TRAP ALERT

City B starts much larger (80,000 vs 50,000), so students may think it will always be ahead. But exponential ALWAYS wins in the long run.

KEY INSIGHT

This is the fundamental principle: exponential growth eventually surpasses any linear growth, regardless of initial values.

FAST METHOD

Exponential beats linear eventually. Always. That is the key principle.

VERDICT

Hand-solve. Pure concept — 5 seconds.

Q21

Hard

Answer: B) 4.4%

Step-by-Step Solution:

$$N(t) = N_0(0.8)^{t/5}.$$

$$\text{After 1 year: } N(1) = N_0(0.8)^{1/5}.$$

$$(0.8)^{0.2} = 0.9564\dots$$

$$\text{Annual retention} = 0.9564, \text{ so annual decay rate} = 1 - 0.9564 = 0.0436 = 4.36\%.$$

Approximately 4.4%.

Why Other Options Are Wrong:

A) 4% — rounds too aggressively or uses a different method.

C) 16% — uses $1 - 0.84$ or makes arithmetic error.

D) 20% — uses the 5-year decay rate (which is 20% over 5 years, not per year).

TRAP ALERT

Option D (20%) is the biggest trap. 20% is the decay over 5 years, not 1 year. To find the annual rate, take the 5th root: $(0.8)^{1/5}$.

KEY INSIGHT

To convert a decay rate from one time period to another, use the appropriate root: annual rate from 5-year rate means taking the 5th root of the retention factor.

FAST METHOD

$$\text{Annual factor} = (0.8)^{0.2}. \text{ Use Desmos: } 0.8^{(1/5)} = 0.9565. \text{ Decay} = 4.35\%.$$

VERDICT

Desmos is faster for computing fractional exponents. 10 seconds.

Q22

Hard

Answer: 6

Step-by-Step Solution:

$$f(1) = ab = 18.$$

$$f(4) = ab^4 = 486.$$

Divide: $ab^4/(ab) = 486/18$.

$$b^3 = 27.$$

$$b = 3.$$

From $f(1) = ab = 18$: $a(3) = 18$, so $a = 6$.

TRAP ALERT

Grid-in: students might solve for b and forget to find a , or they might confuse $f(4)/f(1) = b^3$ (not b^4).

KEY INSIGHT

Dividing two exponential equations eliminates a and isolates a power of b .

FAST METHOD

$$f(4)/f(1) = b^3 = 27 \rightarrow b = 3 \rightarrow a = 18/3 = 6.$$

VERDICT

Hand-solve. Division method — 15 seconds.

Q23

Hard

Answer: B) $g(x)$ eventually grows much faster

Step-by-Step Solution:

$f(x) = 100 + 25x$ is linear: grows by 25 per unit of x .

$g(x) = 100(1.25)^x$ is exponential: grows by 25% per unit of x .

For large x , exponential growth dominates linear growth.

$g(x)$ will eventually far exceed $f(x)$.

Why Other Options Are Wrong:

A) $f(x)$ has a larger initial rate only at $x=0$ where both grow by 25. Beyond that, $g(x)$ accelerates.

C) They do not grow at the same rate — exponential accelerates.

D) $f(x)$ is linear and has no horizontal asymptote.

TRAP ALERT

At $x=0$, both functions equal 100 and both initially "grow by 25." Students might think they are equivalent, but 25% of a growing number eventually dwarfs a fixed 25.

KEY INSIGHT

Exponential growth rate increases over time (since it is a percentage of a growing amount), while linear growth rate stays constant. This is why exponential always wins.

FAST METHOD

Exponential always beats linear for large x . Done.

VERDICT

Hand-solve. Conceptual reasoning — 5 seconds.

Q24

Hard

Answer: A) Geometric series sum

Step-by-Step Solution:

Hour 1: 50 shares. Hour 2: $50(0.7)$. Hour 3: $50(0.7)^2$ Hour 6: $50(0.7)^5$.

Total = $50 + 50(0.7) + 50(0.7)^2 + \dots + 50(0.7)^5$.

This is a geometric series with first term 50, ratio 0.7, and 6 terms.

Sum = $50(1 - 0.7^6) / (1 - 0.7)$.

Why Other Options Are Wrong:

B) This gives only the shares in the 6th hour, not the total.

C) This is a linear model, not geometric.

D) This is just multiplication, not a proper sum.

TRAP ALERT

The question asks for TOTAL shares (sum of all hours), not just the shares in hour 6. Option B gives only the 6th hour's shares.

KEY INSIGHT

When each term is a constant ratio of the previous, the total is a geometric series: $S = a(1 - r^n)/(1 - r)$.

FAST METHOD

Recognize geometric series pattern. Apply the formula directly.

VERDICT

Hand-solve if you know the formula. Desmos can compute the numerical value if needed.

Q25

Hard

Answer: C) $x = 4$

Step-by-Step Solution:

$f(x) = 2^x$ and $g(x) = 4x$.

Test each option:

$x = 1$: $f = 2^1 = 2$. $g = 4(1) = 4$. Not equal.

$x = 2$: $f = 2^2 = 4$. $g = 4(2) = 8$. Not equal.

$x = 4$: $f = 2^4 = 16$. $g = 4(4) = 16$. Equal!

$x = 8$: $f = 2^8 = 256$. $g = 4(8) = 32$. Not equal (exponential far exceeds linear).

Why Other Options Are Wrong:

A) $x = 1$: 2 is not equal to 4.

B) $x = 2$: 4 is not equal to 8.

D) $x = 8$: 256 is far greater than 32 — the exponential has pulled far ahead.

TRAP ALERT

At $x = 8$, the exponential is massively larger. Once 2^x overtakes $4x$, it never comes back. Students who guess large values may pick D.

KEY INSIGHT

Testing integer values is the fastest approach for equations like $2^x = 4x$. Start with the answer choices.

FAST METHOD

Plug in each option: only $x = 4$ gives equal values ($16 = 16$).

VERDICT

Hand-solve. Test each option — 15 seconds.

TOPIC SUMMARY

Core Concept

Linear growth adds a constant amount each period. Exponential growth multiplies by a constant factor each period. The DSAT tests whether you can identify, model, interpret, and compare these two types.

Key Formulas

Linear: $f(t) = \text{initial} + \text{rate} \times t$.

Exponential growth: $f(t) = \text{initial} \times (1 + r)^t$.

Exponential decay: $f(t) = \text{initial} \times (1 - r)^t$.

Doubling time: $f(t) = \text{initial} \times 2^{(t/\text{period})}$.

Half-life: $f(t) = \text{initial} \times (0.5)^{(t/\text{half-life})}$.

How to Identify the Type

Constant differences between consecutive terms = linear.

Constant ratios between consecutive terms = exponential.

Most Common DSAT Question Types

1. Identify linear vs. exponential from a table or description.
2. Write the function given a word problem.
3. Interpret the initial value and growth rate from a function.
4. Find when exponential surpasses linear (intersection).
5. Decay problems: half-life, depreciation, radioactive decay.

Top 3 Traps to Avoid

1. Using the decay RATE (r) instead of the retention factor ($1 - r$) as the base.
2. Putting the period in the wrong place in the exponent (t/p , not pt).
3. Confusing exponential functions $f(x) = a(b)^x$ to the x with power functions $f(x) = ax^n$ to the n .

Desmos vs. Hand-Solve Summary

Identification and interpretation questions: hand-solve (60% of questions). Intersection/crossover problems and fractional exponents: Desmos saves significant time (40% of questions).